

Physical AI Faces One Critical Chokepoint, And Here's How To Profit



BY TYLER DURDEN

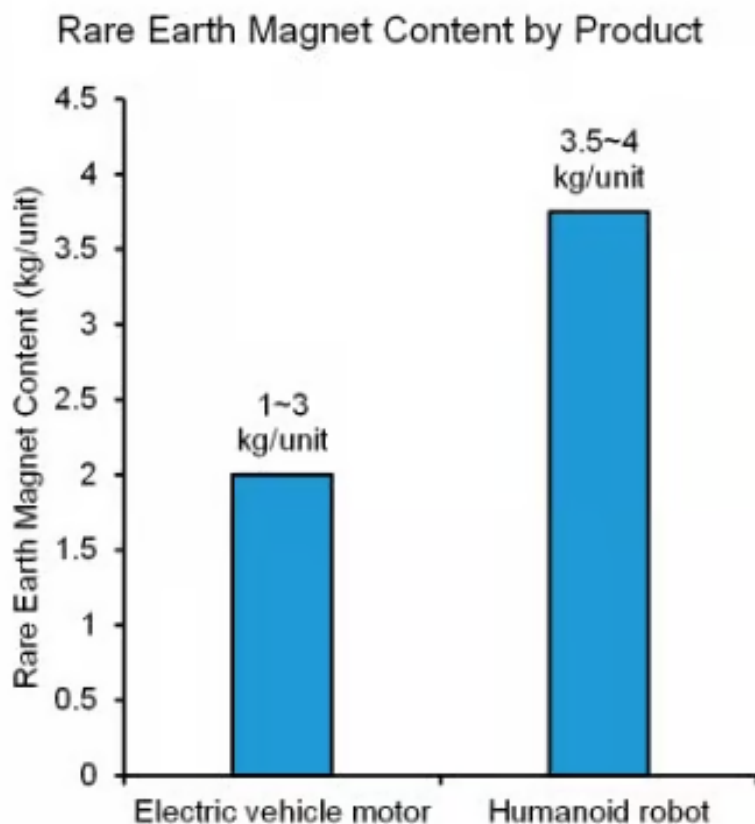
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The **rise of physical AI is already underway**, with humanoid robots entering factories, warehouses, and other industrial environments, and soon the [modern battlefield](#).

Global shipments of humanoid robots are expected to [accelerate in 2027](#), but the **broader robotics revolution faces one critical chokepoint**: *access to the rare-earth magnets that power high-performance motors and actuators. China controls most of that supply chain.*

Bernstein analyst Dien Wang told clients Monday that humanoid robots require an estimated 3.5 to 4 kilograms of rare-earth magnets per unit, compared with roughly 1 to 3 kilograms for an EV motor.

EXHIBIT 3: **Rare-earth magnet content per EV motor and humanoid robot**

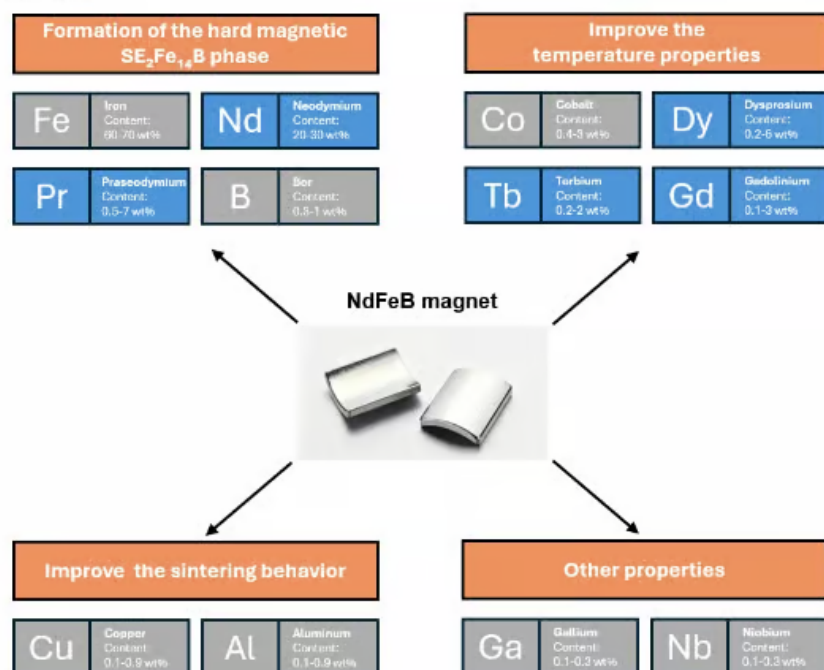


Source: Rare Earth Exchanges, IEA, China Northern Rare Earth, Bernstein analysis

"A **humanoid robot can consume more than twice as much magnetic material as an EV motor** (Exhibit 3), highlighting the growing strategic importance of rare-earth magnets. Beyond well-known neodymium (Nd) and samarium (Sm), rare-earth elements such as dysprosium (Dy), terbium (Tb), and praseodymium (Pr) are also critical for magnetic performance (Exhibit 4, Exhibit 5). **We therefore see the entire rare-earth magnet value chain as strategically important,**" Wang said.

Rare-earth magnets (NdFeB) are composed of multiple raw materials, including various rare-earth elements.

EXHIBIT 4: **Rare-earth magnets (NdFeB) are composed of multiple raw materials, including various rare earth elements**



Note: Blue boxes indicate rare earth elements (REEs).
Source: Deutsche Rohstoffagentur, Bernstein analysis

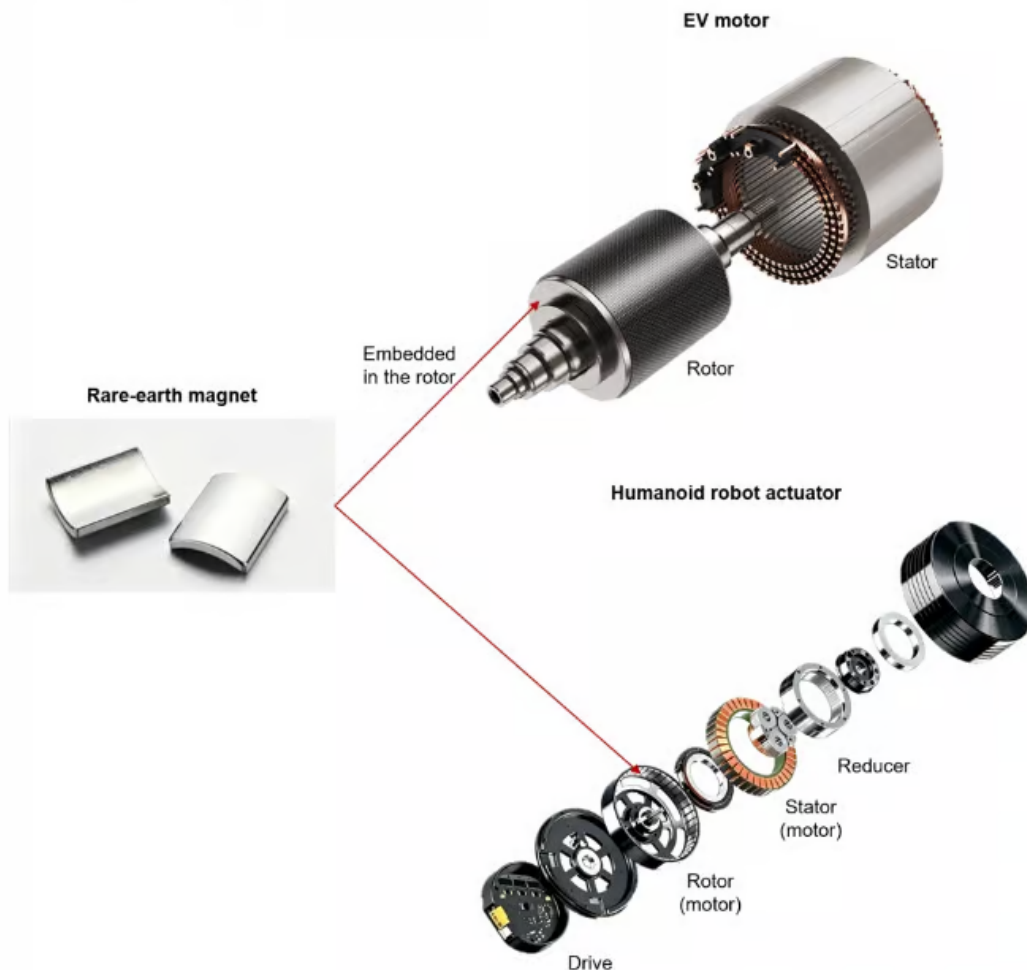
EXHIBIT 5: **Rare-earth elements are critical to magnet performance, e.g. maximum operating temperature**

Max operating temp. (°C)	Approx. Dy content (wt. %)	Approx. Nd+Pr content (wt. %)	Example applications
80	<0.5%	29.5%	Toys, games, advertising, etc.
100	1.4%	28.6%	Hard disk drives, CD/DVD, MRI machines, sensors, refrigeration, etc.
120	2.8%	27.2%	Gauges, hysteresis clutch, magnetic separation, etc.
150	4.2%	25.8%	Wind power generators, industrial motors, electric bicycles, magnetic braking, general automotive applications, etc.
180	6.5%	24.5%	Commercial and industrial generators, wave guides, undulators, etc.
200/220	8.5% - 11%	19% - 21.5%	EV motors, hybrid vehicle motors, high temp. motors and generators, etc.

Source: U.S. Department of Energy, Bernstein analysis

The magnets provide the torque density, compact size, and heat resistance needed for **high-performance motors and robotic actuators.**

EXHIBIT 1: **Rare-earth magnets: Critical to EV motors and humanoid robot actuators**



Source: Inovance Automotive, CubeMars, Bernstein analysis

Wang outlined that **China's dominance in rare-earth magnets is troubling in an "increasingly multipolar world."** He said there are only three conclusions to draw from this:

1. *Chinese EV (e.g. BYD) and humanoid OEMs could benefit from an improved competitive position overseas.*
2. *Chinese robotic actuator (e.g. Tuopu, Sanhua) and EV-motor suppliers (e.g. Inovance) could gain share globally, as rival supply chains remain more vulnerable to rare-earth magnet disruptions.*
3. *Non-Chinese rare-earth magnet players (Exhibit 17) could benefit from growing requests for supply-chain diversification.*

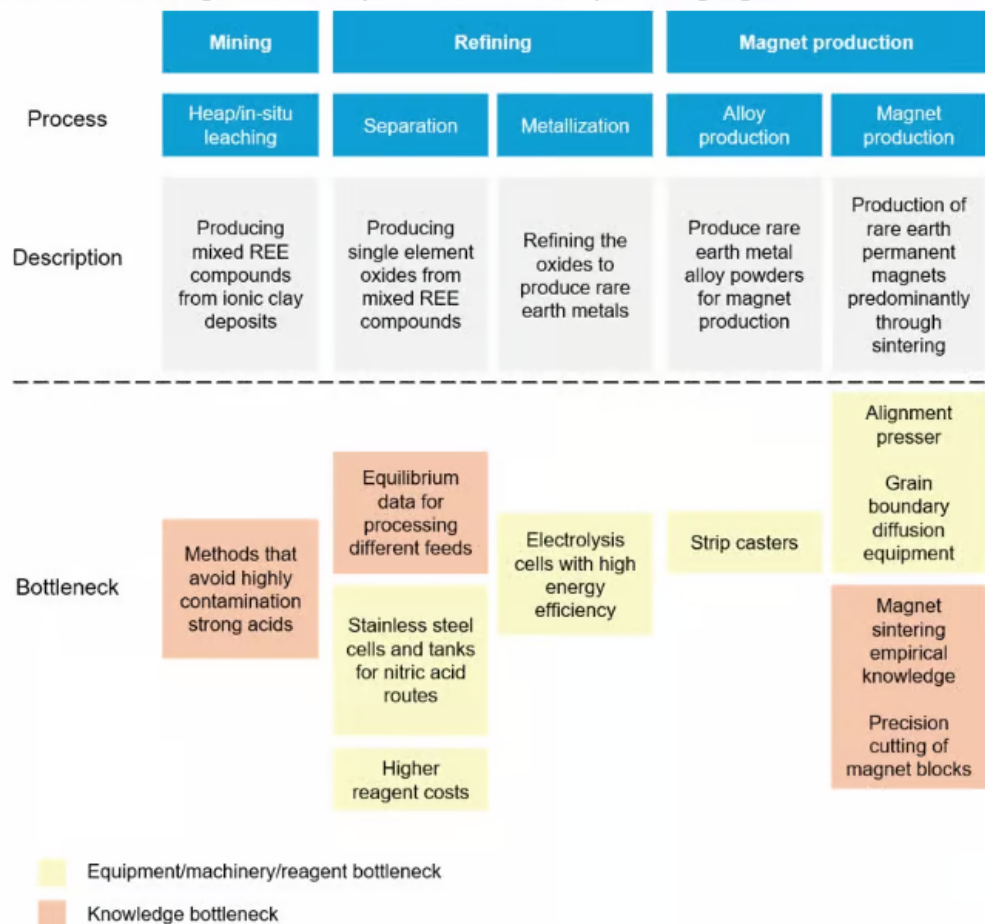


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Rare-earth magnet value chain spans a series of technically demanding stages

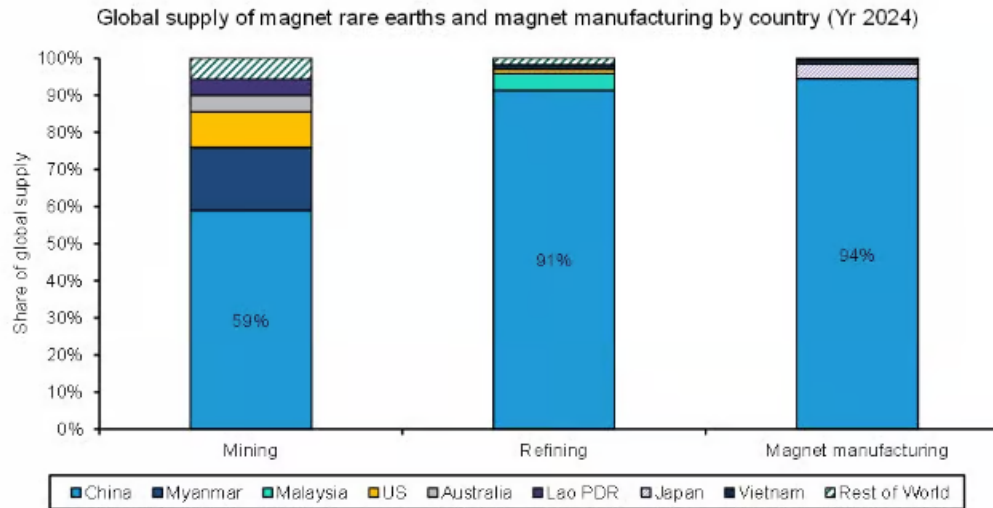
EXHIBIT 6: **Rare-earth magnet value chain spans a series of technically demanding stages**



Source: IEA, Bernstein analysis

China controls about 59% of global magnet rare-earth mining, 91% of refining and 94% of magnet manufacturing, according to the analyst.

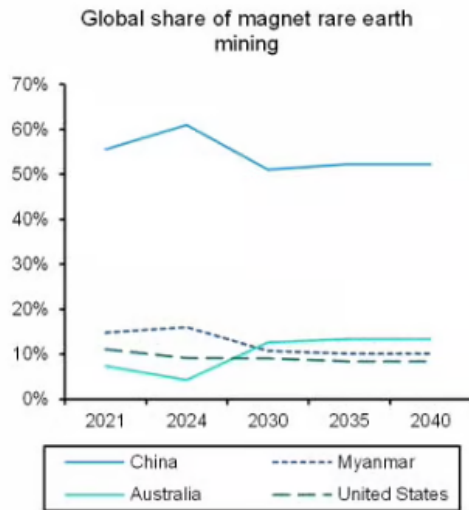
EXHIBIT 7: **China dominates the value chain of rare-earth magnet**



Source: IEA, Bernstein analysis

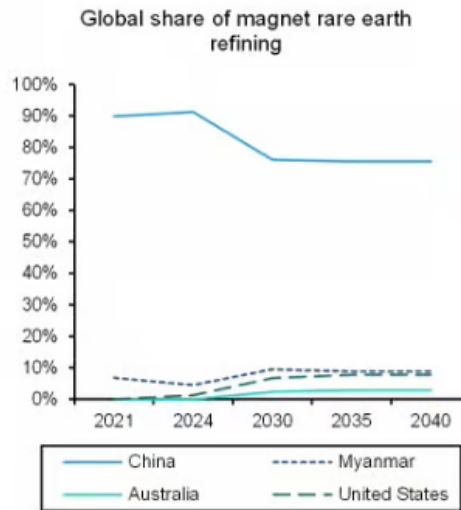
The International Energy Agency expects **China to retain more than half of mining and over 75% of refining through 2040**, while recycling is unlikely to meaningfully reduce near-term supply risks.

EXHIBIT 8: **According to IEA estimates, China will maintain a global share of >50% in magnet rare earth mining**



Note: IEA's projections are based on the pipeline of operating and announced mining and refining projects across countries.
Source: IEA, Bernstein analysis

EXHIBIT 9: **According to IEA estimates, China will maintain a global share of >75% in magnet rare earth refining**



Note: IEA's projections are based on the pipeline of operating and announced mining and refining projects across countries.
Source: IEA, Bernstein analysis

Beijing has weaponized its dominance of rare earths as a geopolitical tool against the US. China imposed controls on seven medium and heavy rare-earth products in 2025, extended licensing requirements to some overseas transactions, and restricted exports to dozens of US, European, and Japanese companies.

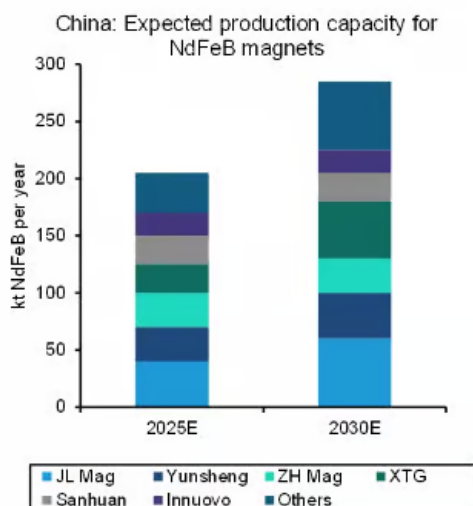
Those 2025 measures caused an immediate, though temporary, collapse in Chinese magnet exports.

EXHIBIT 15: **China has been strengthening controls across the rare earth supply chain**



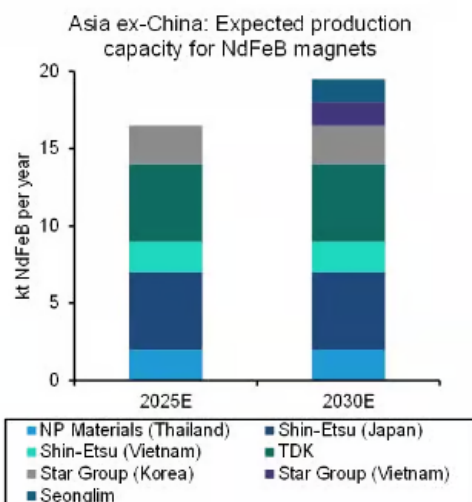
China/Asia Production of NdFeB magnets

EXHIBIT 10: China: Production capacity of NdFeB magnets



Note: The forecast is made by IEA. The list of projects is indicative of future magnet production capacity based on current announcements, it is not meant to be exhaustive.
Source: IEA, Bernstein analysis

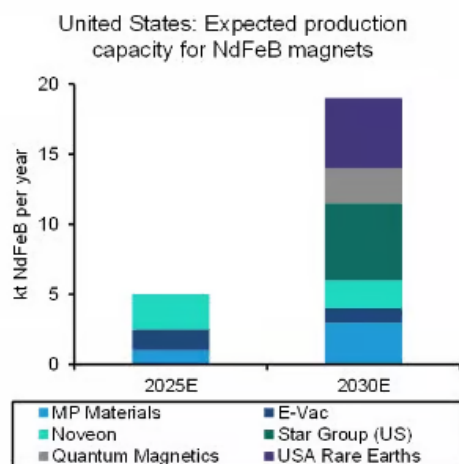
EXHIBIT 11: Asia ex-China: Production capacity of NdFeB magnets



Note: The forecast is made by IEA. The list of projects is indicative of future magnet production capacity based on current announcements, it is not meant to be exhaustive.
Source: IEA, Bernstein analysis

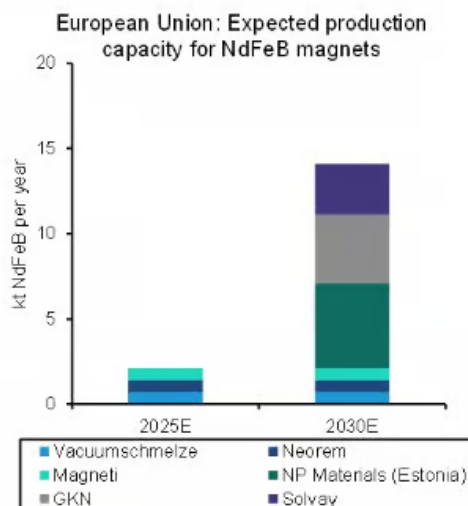
Comparing with the US and Europe...

EXHIBIT 12: United States: Production capacity of NdFeB magnets



Note: The forecast is made by IEA. The list of projects is indicative of future magnet production capacity based on current announcements, it is not meant to be exhaustive.
Source: IEA, Bernstein analysis

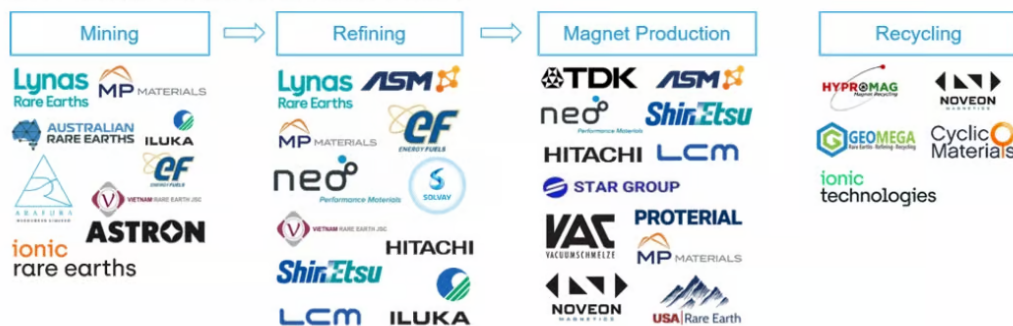
EXHIBIT 13: European Union: Production capacity of NdFeB magnets



Note: The forecast is made by IEA. The list of projects is indicative of future magnet production capacity based on current announcements, it is not meant to be exhaustive.
Source: IEA, Bernstein analysis

Here's how to profit across the entire magnet supply chain ex-China:

EXHIBIT 17: Rare-earth magnet players outside China



Note: Solvay SA (SOLB.BB) is covered by Bernstein European Chemicals team and others are not listed or covered by Bernstein.
Source: Neo Performance Materials, company websites, IEA, Bernstein analysis

"As Physical AI emerges as a strategic battleground among leading nations, control of critical chokepoints could ultimately determine who gains the upper hand," Wang noted.

China may already hold a structural advantage in humanoid robot development, supported by its massive manufacturing ecosystem and dominance of the rare-earth supply chain. On Monday, Hugging Face's CEO told CNBC that China is winning the AI race and increasingly dominating open-source models. **Taken together, these signals suggest the West's lead in AI might be eroding against China.**

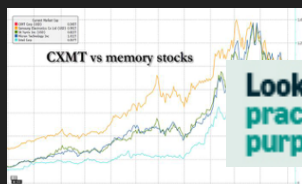
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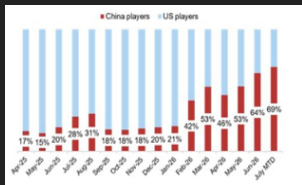
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